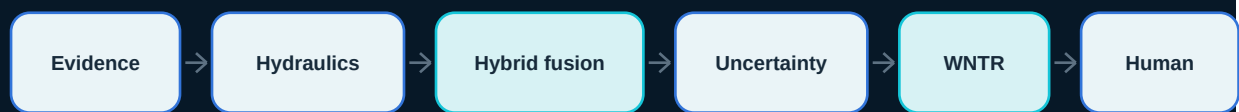


HydroSwarm

Offline Neuro-Hydraulic Incident Intelligence

Physics-first source localization, active evidence selection, and exact response verification



Typed evidence and explicit uncertainty; no autonomous control path

REVERIE HACKS 2026 / SOFTWARE DEVELOPMENT

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Safety boundary

Research decision support only. No chemistry identification, no autonomous control, and no field action without qualified human approval.

Abstract

Drinking-water quality alerts are difficult to interpret because observations are sparse, hydraulic direction changes over time, sensors fail, and interventions can create pressure or service failures. HydroSwarm is an offline decision-support system that combines hydraulic state reconciliation, time-directed graph screening, Bayesian source signatures, a graph and time-series neural model, calibrated uncertainty, active sampling, and exact WNTR verification. It produces typed evidence and operational alternatives rather than autonomous commands. A frozen WNTR regression scenario demonstrates the complete loop: four initially plausible sources, a measured one-bit sample selection, candidate contraction, rejection of an unsafe closure, verification of a safer flush alternative, comparison against no response, a human approval pause, and deterministic replay. Results are simulation measurements and are not claims of field performance.

1. Problem and motivation

A single abnormal observation rarely identifies where an incident began. Water moves through a network whose effective direction depends on demand, storage, pumps, valves, and time. Several upstream locations can produce similar observations, while a frozen or mis-scaled sensor can imitate a network event. Field sampling is slow and costly. Response actions also couple public-health objectives to hydraulic constraints: isolation or flushing can reduce modeled exposure yet cause low pressure or unserved demand.

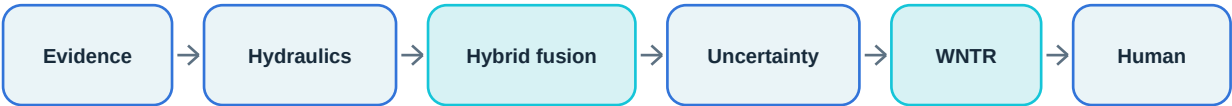
HydroSwarm addresses the interval between detection and an operator decision. It is built for incident commanders, hydraulic engineers, water-quality staff, and sampling teams. It does not identify chemistry, certify potability, issue public-health advice, connect to SCADA, or execute infrastructure changes.

2. Related tools and methods

WNTR provides the authoritative EPANET-based hydraulic and quality simulation layer. NetworkX supports explicit graph algorithms. The learned architecture is informed by GraphGPS-style local and global processing, Perceiver-style latent attention, and parameter-efficient adapters. Split conformal prediction supplies empirical candidate-set coverage, while energy and disagreement signals contribute to out-of-distribution detection. Expected information gain connects uncertainty to the next field sample.

The central design decision is hybrid rather than purely neural: a source-signature model remains inspectable and operational when no checkpoint is present, and exact simulation is the final authority for response-plan status.

3. System architecture



Typed evidence and explicit uncertainty; no autonomous control path

Telemetry and samples first enter state reconciliation. A hydraulic run produces the time-specific directed graph. Classical feasibility and signature likelihood operate in parallel with HydroCore. Fusion trust changes with sensor health, disagreement, calibration validity, and OOD evidence. The resulting posterior controls either active sampling or plan generation. Every plan passes deterministic constraints and exact WNTR consequence analysis before it can be labeled VERIFIED. A person must then approve it; HydroSwarm has no action-execution interface.

FastAPI serves typed JSON, WebSocket updates, background-job progress, and the built React console from one local process. SQLite persists networks, incidents, evidence, posterior history, jobs, plans, approvals, and a hash-chained event ledger. Runtime modes degrade from full hybrid through degraded hybrid and classical safe to simulation-only.

4. Classical inference

State reconciliation adjusts demand multipliers, tank state, and pump or valve status within bounded uncertainty and reports residual mismatch. The dynamic graph orients links using simulated flow at the relevant time; near-zero and reversed flows remain explicit. Feasibility screening removes sources inconsistent with directed travel time and sample timing.

The source-signature service enumerates source node, start time, duration, strength, and demand regime. Exact quality simulations produce sensor-time response tensors. Compressed NPZ artifacts carry a complete cache key and checksum; mismatch or corruption forces an exact rebuild. A Gaussian residual likelihood and prior produce hypothesis probabilities, which are aggregated to nodes and connected candidate regions. Explanations include residual magnitude and physical exclusion reasons.

5. HydroCore architecture

HydroCore receives canonical 19-dimensional node features, 13-dimensional edge features, temporal sensor histories, observation masks, and sensor-quality features. Local edge-aware graph processing captures adjacency and direction. A fixed latent bottleneck provides global attention at bounded cost. RMS normalization, SiLU activations, residual connections, and dropout support stable training.

Semantic heads predict source logits, incident timing and strength, sensor health, sampling value, typed plan actions, consequence residuals, and fixed explanation intents. Small, medium, and large configurations contain 4,040,645, 12,401,861, and 24,415,397 parameters. HydroMono and no-adapter variants support ablation. Source logits learn a residual over the classical signature prior and exclude reservoirs and tanks with a typed candidate mask. The promoted S checkpoint and calibration artifact are hash-validated; missing or incompatible assets fail closed to classical mode.

6. Specialist agents and control

HydroSentinel validates evidence, assesses sensor versus network events, and localizes a source region. HydroScout ranks accessible, non-duplicate candidates by expected posterior information gain, likely region reduction, detection probability, delay, cost, and redundancy. HydroStrategist creates diverse bounded templates including no response, flushing, isolation, valve and pump adjustments, monitoring, and operator-reviewed information summaries. It does not issue public notifications or emergency alerts. HydroVerifier alone can promote a plan after exact simulation.

An 18-state deterministic controller owns permissions, timeouts, retry budgets, and human checkpoints. Malformed agent output is rejected. Sample arrival triggers a new posterior revision rather than mutating history. Idempotency and replay tests cover repeated events and recovery.

7. Data generation and governance

The WNTR scenario generator randomizes source, timing, duration, strength, demand, tank level, roughness, outages, sensor layout, noise, drift, freezing, communication loss, jitter, quantization, unit mismatch, and flow reversal. Curriculum stages progress from clean through operational, degraded, distribution-shift, and adversarial conditions.

Network-disjoint split ownership is assigned before simulation. Manifests record seeds, network and artifact hashes, simulator and generator versions, physical settings, sensor layout, corruption settings, replay hash, and provenance. Validators enforce finite values, aligned time axes, correct missing-value masks, replay consistency, and seed-family separation. Calibration and final test partitions are never used for model fitting.

The measured learning-v1 corpus contains 1,320 scenarios: 800 train, 160 validation, 160 calibration, and 200 test. Five curriculum stages each contain 264 incidents and four source nodes each contain 330. Four hydraulic regimes are used for training and one regime is withheld from training, signature fitting, and calibration. All regimes share one topology; this is hydraulic-shift evidence, not unseen-topology transfer.

8. Training

Training uses governed JSONL trajectories and PyTorch. Multitask semantic losses cover localization, event parameters, sensor health, sampling, actions, consequences, and explanation intents. AdamW, warmup scheduling, gradient accumulation, clipping, mixed precision where supported, early stopping, task-weight logging, optional GradNorm, and optional PCGrad are available. Runs record the resolved configuration, git state, dataset manifest hash, seeds, environment, progress, failures, and resource guards.

Checkpoints are safetensors with resume state and checksums. Training can resume after a bounded interruption. A language decoder, when used, is trained against fixed evidence intents with the operational latent detached so fluent text cannot alter scientific heads.

HydroCore-S and HydroMono-S each completed eight epochs and 600 optimizer steps on CPU in 385 and 397 seconds respectively. HydroCore-M completed 17 epochs and 1,500 optimizer steps under a predefined 2,400-second budget. Checkpoint selection used validation only; the original test hashes remained frozen. Profile losses ignore architecturally reserved classes outside the governed 4/3/3 label spaces and add an ordinal-aware penalty. Published records include curves, task losses, configuration, environment, dataset hash, and safe candidate weights without optimizer state.

9. Uncertainty, OOD, and abstention

Split and Mondrian conformal artifacts are fit only on held-out calibration examples and versioned against model, feature schema, and dataset manifest hashes. Reports include coverage, mean set size, expected calibration error, and condition/network breakdowns. Invalid artifacts fail closed.

OOD assessment combines feature-range departure, missingness, energy score, topology-hash novelty, classical/neural disagreement, and physical residuals. An unseen validated-set topology forces at least CAUTION. Severe OOD or invalid calibration suppresses confident planning. Dynamic trust favors classical evidence under sensor/model conflict and neural residual correction only within validated conditions. Abstention is visible to the operator and changes the controller path.

10. Plan verification and consequences

Plans are typed action sequences with targets, values, units, timing, constraints, and rationale references. Deterministic checks limit action count, targets, ranges, conflicts, and exact-simulation budget. WNTR evaluates pressure, demand delivery, tank behavior, quality transport, exposure mass and volume, population and pipe-extent proxies, pump energy, and service consequences. Timeouts, instability, missing output, or partial runs are rejections.

Plans are compared with a separately simulated no-response branch. Multi-objective ranking reports Pareto status and regret instead of hiding tradeoffs in one scalar. VERIFIED means only that all configured simulation constraints passed; APPROVED is a separate recorded human decision and never triggers physical execution.

11. Evaluation protocol

The evaluation framework measures localization top-k and true-source probability, candidate contraction, entropy reduction and information gain, unsafe rejection, safe acceptance, approval pause, deterministic replay, exposure reduction, latency, Python allocation peak, and cache behavior. Repeated seeded runs produce confidence intervals. Baselines and ablations include uniform localization, no active sampling, no exact verifier, and no cache. Neural variants are evaluated only when compatible trained checkpoints are supplied.

The frozen golden network, scenario, and manifest are checksummed. All displayed outcomes are regenerated through WNTR. Runtime is excluded from the reproducibility hash, while scientific inputs and results remain included.

12. Measured results

Measured metric	Value	Interpretation
Top-1 localization	1	Frozen WNTR regression
True-source probability	0.9941	Frozen WNTR regression
Entropy reduction	1.948	Frozen WNTR regression
Modeled exposure reduction	1.472e+04	Frozen WNTR regression
Golden latency	0.9801	Frozen WNTR regression
Logical cache hit rate	0.6667	Frozen WNTR regression

The checked-in regression report passes its promotion gate. A uniform four-node prior contracts to the true J2 source at approximately 99.4 percent after sampling J2. The sample has 1.0 bit measured information gain. WNTR rejects a sole-feeder closure

for pressure below the minimum and verifies a J4 flush alternative with full modeled service. Separate quality runs estimate 2,541,416 mg no-response consumption and 2,526,693 mg for the verified alternative, a reduction of 14,723 mg. The controller pauses for both sampling and human approval and completes deterministic hash-chain replay.

These values are a compact regression proof, not a population estimate and not evidence of real-world accuracy. The authoritative machine-readable results are included alongside this report.

A governed learned benchmark evaluates 200 withheld hydraulic-regime incidents. The classical signature baseline reaches 91.5 percent top-1 accuracy (95 percent bootstrap CI 87.5-95.0). HydroCore-S and equal-budget HydroMono-S each reach 94.5 percent. S hybrid fusion reaches 96.0 percent (93.0-98.5), a 4.5-point improvement over classical. The budget-complete M candidate reaches 94.5 percent neural-only and 94.0 percent hybrid, with 23.93 ms mean scenario latency versus 8.94 ms for S. M therefore fails its operational promotion gate and S remains the runtime default.

M conformal calibration, fit only on 160 calibration cases, reaches 91.0 percent test coverage, mean set size 0.93, and ECE 0.0367. M start-time and strength-bin accuracy improve from 20.5 to 27.0 percent and 34.0 to 45.5 percent; duration falls from 42.5 to 35.5 percent. These heads remain exploratory.

A separate 70-case evaluation uses a genuinely different seven-junction branched-loop topology without neural fine-tuning. Classical, M neural, and M hybrid top-1 are 35.7, 47.1, and 44.3 percent. Coverage collapses to 27.1 percent with mean set size 0.41. The unseen hash produces CAUTION with novelty 1.0 and suppresses planning. This supports the fail-closed safety story, not a topology-transfer claim.

A separate seeded HydroCore-S FP32 runtime profile uses random weights solely to measure execution behavior. On the current four-thread CPU host, 128-node inference measures 31.00 ms median and 32.34 ms p95 over ten iterations. A 1,000-node canonical model stress case measures 104.33 ms median over five iterations. The 4.04M-parameter safetensors artifact is 16.19 MB, and repeated eager outputs have zero maximum absolute difference. These values do not measure learned accuracy and do not include a 1,000-node WNTR solve.

13. Ablations and failure cases

Removing exact verification fails the promotion gate by design: a proposal cannot become VERIFIED. Disabling the signature cache is actually executed and increases repeated-run latency. Removing active sampling leaves the broad source set unchanged. The final equal-budget HydroMono ablation ties HydroCore-S at 94.5 percent; a specialist-adaptor gain is therefore not established. M does not improve hybrid localization or cost-adjusted performance. HydroCore-L, learned Scout/Strategist, language-gradient, and quantization ablations remain unrun.

Important failure cases include non-identifiable sensor layouts, stale hydraulic state, unknown controls, cache mismatch, unit error, timing jitter, frozen or drifting sensors, simulation timeout, numerical instability, and distribution shift. Each produces either a visible warning, broader uncertainty, fallback, abstention, or rejection rather than a silent success label.

14. Security, reliability, and deployment

The native application binds to loopback, performs no runtime internet calls, accepts no URLs, validates and hashes INP files, limits request and job sizes, and stores data only in its configured directory. The container publishes loopback only, drops capabilities, prevents privilege escalation, uses a read-only root filesystem, and persists a dedicated data volume. Self-test executes real inference, WNTR, SQLite, dependency, resource, port, and frontend checks.

The React interface is API-first and labels its deterministic fixture fallback. It provides a tile-free operational map, topology view, hydraulic charts with text equivalents, timeline replay, evidence-change display, synchronized plan branches, audit and validation views, keyboard navigation, visible focus, reduced motion, and non-color status cues.

15. Limitations

HydroSwarm has not been validated on live utility incidents. Synthetic and reference-network evaluation cannot establish field safety, chemical identity, toxicity, pathogen viability, regulatory compliance, or public-health outcomes. WNTR predictions inherit topology, demand, control, mixing, and sensor-model errors. Population exposure is a proxy unless governed demographic and consumption data are supplied. The API is not an authenticated internet-facing multi-tenant service.

No output authorizes flushing, isolation, equipment changes, notification, or any other field action. Qualified utility and public-health personnel must follow current procedures, laboratory evidence, regulations, and engineering review.

16. Reproducibility

Python dependencies are frozen by uv; the exported runtime set is hash-locked and audited. The React dependency lock is vulnerability-audited. CI covers Windows and Ubuntu, Python tests with branch coverage, Ruff, Pyright, self-test, wheel build, frontend lint, type/build, component accessibility tests, and Playwright. The golden scenario and evaluation scripts write JSON and Markdown outputs with stable scientific hashes.

Reproduction commands are documented in README, INSTALLATION, DATA_GENERATION, and EVALUATION. The final runtime requires no hosted AI service or internet connection.

17. Future work

Next steps require utility partnerships, licensed field telemetry, prospective incident exercises, chemistry-specific transport validation, network-specific conformal calibration, authenticated deployment, operator usability studies, and external scientific review. Optional OpenVINO and INT8 benchmarking, large-network WNTR stress tests, and further sparse attention optimization should follow only after compatible checkpoints and governed calibration data exist.

18. AI assistance

ChatGPT/Codex and Claude/Claude Code were used for implementation assistance, debugging, test generation, documentation review, and architecture critique. The project author selected the architecture, scientific objectives, evaluation methodology, claims, and final implementation, and validated the submitted system. AI services are not runtime dependencies.

19. References

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